

The convenience benefits of the shipping market: Evidence from C3 and C5 FFAs

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ARTICLE INFO

JEL codes:

G15

G17

Keywords:

Forward freight agreements

Convenience yield

Shipping market

Market forecasting

ABSTRACT

Forward Freight Agreements (FFAs) are consequential derivatives for risk hedging in the shipping market. This paper presents the first empirical study of convenience yield in the shipping market with a sample of C3 (Tubarao to Qingdao) and C5 (Western Australia to Qingdao) FFAs from 2013 to 2023. First, single-factor and two-factor models are constructed to analyze the random behavior of freight rates in the shipping market, and the existence of convenience yield is empirically tested. The results provide evidence of the existence of convenience yield in the shipping market, with its stochastic variation being a critical factor in shaping the term structure characteristics of FFAs. Second, an ARMAX model is developed for the freight rate index, and the predictive role of the estimated convenience yield and its difference in spot freight rates is analyzed by operating the two-factor model. The findings reveal a significant predictive effect of convenience yield and its difference on spot freight rates. Last but not least, this study emphasizes the importance of convenience yield as a crucial parameter in the shipping market, particularly regarding the transportation of bulk commodities. Changes in the supply and demand of bulk commodities directly impact the fluctuations in shipping market capacity and convenience yield. This study delivers a new explanatory perspective for price formation in the shipping market and is highly valuable for risk management in this sector.

1. Introduction

The shipping market exhibits apparent volatility, strong seasonality, significant cyclicity, and high capital intensity. Freight rate risk generates uncertain cash flows and unforeseen volatility for ship owners and charterers, underscoring the crucial nature of risk management in the shipping market (Kavussanos & Visvikis, 2006). Shipping derivatives, particularly forward freight agreements (FFAs), are crucial in mitigating market risks, serving as the primary tool for managing freight rate risk in the shipping market. FFAs exhibit heightened sensitivity to new information compared to spot freight rates. Analyzing information transmission between FFAs and spot rates is critically significant for market participants in predicting shipping prices and constructing investment portfolios (Alizadeh, 2013).

Convenience yield in the shipping market is crucial for risk management strategies. Understanding and quantifying convenience yield can provide valuable insights into market conditions, supply-demand dynamics, and future price movements. This information is essential

for shipping companies, charterers, and investors to make informed decisions about hedging strategies, contract negotiations, and investment timing. While convenience yield has been extensively studied in commodity markets (Kaldor, 1939; Pindyck, 2001), its application in the shipping market still needs to be explored. This study aims to bridge this gap by demonstrating how convenience yield can be a powerful tool for risk management in the volatile shipping industry. This paper presents the first empirical study of convenience yield in the shipping market with a sample of C3 (Tubarao to Qingdao) and C5 (Western Australia to Qingdao) FFAs from 2013 to 2023. Compared to the single-factor model, we present a dual-factor model incorporating the mean-reverting process of convenience yield, satisfactorily fitting the FFAs term structure. This suggests that, similar to commodities, the shipping market also exhibits convenience yield, substantially impacting the term structure of shipping derivatives. Furthermore, we construct an ARMAX model by utilizing the convenience yield of the shipping market and its differenced series as exogenous variables. Our findings reveal that the convenience yield significantly predicts spot freight rates in the shipping

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<https://doi.org/10.1016/j.rtbm.2024.101182>

Received 23 January 2024; Received in revised form 3 August 2024; Accepted 15 August 2024

Available online 24 August 2024

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market.

This paper has three momentous implications. Firstly, it contributes to advancing the theory of convenience yield in the shipping market. Existing research needs to pay more attention to the convenience yield in the shipping market. This study demonstrates the importance of the convenience yield as a parameter in the shipping market while also providing economic explanations and mechanisms underlying its implications. Secondly, by utilizing a random behavior model characterized by mean-reversion, we accurately model the term structure of FFAs and identify the stochastic variation of the convenience yield as a critical factor in shaping the nonlinear characteristics of the FFA term structure. Thirdly, convenience yield and their changes predict spot freight rates, essential for risk management and decision-making in the shipping market. Particularly in a market environment with frequent capacity and price fluctuations due to changes in commodity supply and demand, this study provides valuable information and strategic guidance to shipping companies and investors. We discover that the convenience yield in the shipping market contains abundant market information, thus providing fresh theoretical support for the pricing and design of shipping derivatives contracts.

The primary objective of this paper is to develop single-factor and dual-factor stochastic process models for freight prices that incorporate the mean-reverting characteristics of the shipping market (Adland & Cullinane, 2006; Moutzouris & Nomikos, 2020; Sodal, Koekebakker, & Adland, 2014). Only the mean reversion of spot freight rates is considered in the single-factor model. In contrast, the dual-factor model introduces convenience yield as the second state variable and constructs a joint mean-reverting process for convenience yield and spot freight rates. The parameters of the single-factor and dual-factor models are estimated using the Kalman filtering method (Schwartz, 1997). Firstly, empirical results demonstrate the existence of convenience yield in the shipping market. Considering the random process of the convenience yield, the dual-factor model exhibits non-monotonic characteristics in the term structure of FFAs. Secondly, the dual-factor model outperforms the single-factor model regarding overall fit to the term structure, as evidenced by lower fit errors (MAE, MSE), especially for shorter-term monthly contracts and longer-term annual contracts. Therefore, the empirical evidence confirms the presence of convenience yield in the shipping market, and the stochastic process of convenience yield is a primary determinant of the nonlinear properties observed in the term structure of FFAs. Lastly, the transportation of commodities directly influences the convenience yield in the shipping market. Given the cyclical patterns in commodity consumption and transportation demand, the convenience yield in the shipping market displays notable seasonality and cyclicity.

Our second objective is to examine the predictive power of convenience yield on shipping freight rates. To assess the predictive accuracy, we adopt a rolling window methodology to estimate the convenience yield and calculate its first-order difference, capturing the variations in convenience yield due to recent market information. Utilizing the Baltic Capesize Index (BCI) as an indicator to reflect the spot freight market, we construct an ARMAX(1,1) model for the logarithmic returns of the BCI, including the convenience yield and its differenced series as exogenous variables. The model results indicate a significant relationship between the lagged convenience yield in the shipping market and future freight prices. Additionally, the first difference of the convenience yield significantly influences future freight prices, capturing changes resulting from recent market information. Increasing the shipping market's convenience yield indicates decreased idle capacity, thereby driving up freight prices.

This paper builds upon the research conducted by Benth & Koekebakker (Benth & Koekebakker, 2016), who introduced and investigated the concept of convenience yield factor in a stochastic dynamic model for FFAs pricing in the shipping market. In addition, we conduct a more in-depth analysis of the economic implications and mechanisms of convenience yield in the shipping derivatives market based on their

initial findings. Moreover, this study expands the convenience yield concept to markets that lack storage properties. Previous studies by Working (Working, 1949) and Brennan (Brennan, 1976) have proposed that convenience yield is an endogenous outcome of the interplay between supply, demand, and storage decisions in commodity markets. However, recent studies, such as the one conducted by Milonas & Paratsiokas (Milonas & Paratsiokas, 2017) on non-storable electricity markets, have demonstrated the existence of convenience yield in markets without storage properties, where the value of fuel (including convenience yield) is transmitted to power output through the production process. Omura et al. (Omura, Todorova, Li, & Chung, 2015) examined the relationship between net convenience yield in power markets dominated by hydroelectricity and the overall water level in the system's reservoirs, finding that the net convenience yield decreases as the total water level increases. Combining the dynamic correlation between the shipping market and commodity market (Adewuyi, Adeleke, Tiwari, & Abakah, 2023; Gu, Chen, & Lien, 2019; Poulakidas & Joutz, 2009), we find evidence that convenience yield can be transmitted from commodity markets to the shipping market, thus extending the study of convenience yield to the non-storable shipping market.

Our study is also relevant to the research on freight rate forecasting in the shipping market. Freight rate forecasting is an essential topic in risk management in the shipping market (Gavriliadis, Kambouroudis, Tsakou, & Tsouknidis, 2018; Munim & Schramm, 2017; Tsioumas, Papadimitriou, Smirlis, & Zahran, 2017). This paper investigates the predictive power of market information captured in the shipping market's convenience yield for spot freight rates, which is significant for risk management in the shipping market.

2. Models

This paper employs classical models that depict the stochastic nature of commodity prices, including single-factor and two-factor models, to analyze the stochastic behavior of shipping market prices. To examine convenience gains in the shipping derivatives market, we introduce them as the second state variable in the two-factor model. Following the works of Schwartz (Schwartz, 1997), Ewald et al. (Ewald, Nawar, Ouyang, & Siu, 2016), and Tretheway (Tretheway, 2004), we developed a two-factor model incorporating joint mean reversion for spot freight rates and convenience yields. Previous studies by Groll et al. (Groll, López-Cabrera, & Meyer-Brandis, 2016) and Dai et al. (Dai, Xiong, Zhang, & Zhou, 2022) have applied the two-factor model to studies of futures and derivatives markets. Building on this foundation, we adapt the model to the study of bulk commodities in the shipping market based on the characteristics of FFA data.

2.1. One-factor model

Previous studies (Adland & Cullinane, 2006; Tretheway, 2004) indicate that competitive freight markets must return to long-term price levels. Similar studies (Benth & Koekebakker, 2016; Benth, Koekebakker, & Taib, 2015; Sodal, Koekebakker, & Aadland, 2008) usually employ a stochastic process with mean reversion to portray spot freight rates. Therefore, this paper assumes that spot freight rates follow the following stochastic process

$$dS = \kappa_s(\mu - \ln S)Sdt + \sigma_s Sdz \quad (1)$$

where $\kappa_s, \sigma_s > 0$, S is the spot freight price, κ_s is the mean reversion rate of spot freight rates, μ is the long-run mean of the logarithmic price, σ_s is the volatility of the spot freight rate, dz denotes an increment of the standard Brownian motion associated with spot freight rates. Giving the definition $X = \ln S$ and assuming X obeying the normal distribution, applying Ito's Lemma yields:

$$dX = \kappa_s(\alpha_s - X)dt + \sigma_s dz \quad (2)$$

Table 1
Descriptive Statistics of C3 FFAs.

Contracts	# of Obs.	Min value	Max value	Average value	Standard deviation	Kurtosis	Skewness
M0	1908	5.48	46.25	18.41	6.97	3.56	0.61
M1	1908	5.55	41.62	18.54	7.01	3.02	0.56
M2	1908	5.85	38.00	18.45	6.93	2.85	0.55
Q1	1908	6.70	37.50	18.19	6.68	2.89	0.58
Y1	1908	7.95	25.75	16.83	4.24	2.04	-0.11

where $\alpha_s = \mu - \frac{\sigma_s^2}{2\kappa_s}$. Consider the market price of risk λ_s . Let $\alpha_s^* = \alpha_s - \lambda_s$, then α_s^* denotes the long-run average of the log price considering market risk. Under the risk-neutral measure Q , formula (2) can be rewritten as:

$$dX = \kappa_s(\alpha_s^* - X)dt + \sigma_s dz^* \quad (3)$$

According to classical expectation theory, in the shipping market, the forward price with maturity T converges to the expected price at the moment T , i.e., $F(S, T) = E^Q[S(T)]$, $S(T) = \exp(X)$. Therefore, we have:

$$F(S, T) = E^Q[S(T)] = \exp\left(\tilde{\mu}_X(T) + \frac{1}{2}\sigma_X^2(T)\right) \quad (4)$$

According to formula (3), the conditional mean and variance of the logarithm of the spot freight rate can be determined as:

$$\begin{cases} \tilde{\mu}_X(T) = \exp(-\kappa_s T)X_0 + (1 - \exp(-\kappa_s T))\alpha_s^* \\ \sigma_X^2(T) = \frac{\sigma_s^2}{2\kappa_s}(1 - \exp(-2\kappa_s T)) \end{cases} \quad (5)$$

By substituting formula (5) into formula (4) and taking the logarithm of both sides, we obtain the closed-form solution for the logarithm of the forward price:

$$\ln F(S, T) = \exp(-\kappa_s T)\ln S + (1 - \exp(-\kappa_s T))\alpha_s^* + \frac{\sigma_s^2}{4\kappa_s}(1 - \exp(-2\kappa_s T)) \quad (6)$$

2.2. Dual-factor model

Convenience yield is the classical theoretical foundation for studying price behavior in the commodity market. Shipping needs storage capability as a transport service, and more research focuses on the convenience yield in the shipping market. Benth and Koekebakker (Benth & Koekebakker, 2016) proposed and employed a stochastic dynamic model to examine the pricing of FFAs in the shipping derivatives market, introducing and anticipating the notion of convenience yield factors in the shipping industry. However, thus far, there needs to be more extensive research on the economic implications and mechanisms of convenience yield in the shipping derivatives market. At least two reasons underscore the importance of studying the convenience yield in the shipping market.

Firstly, convenience yield can be transmitted from the commodity market to the shipping market. Research has demonstrated that, despite electricity being non-storable, convenience yield exists within its market (Milonas & Paratsiokas, 2017). The fluctuations in raw material inventories consumed in the electricity market facilitate the transfer of convenience yield from the commodity market through production. Shipping accounts for approximately 80% of commodity trading volume (Gavriilidis et al., 2018), establishing a natural link between the shipping and commodity markets. Studies conducted by Angelopoulos et al.

(Angelopoulos, Sahoo, & Visvikis, 2020) and Bandyopadhyay & Rajib (Bandyopadhyay & Rajib, 2023) have revealed significant cross-market spillover effects and dynamic correlations between the shipping and commodity markets. Furthermore, the seasonal changes observed in the shipping market are influenced by commodities' seasonal demand and supply patterns. Therefore, it is reasonable to posit that convenience yield in the commodity market is transmitted to the shipping market.

Secondly, evidence suggests the existence of "convenience" in the spot shipping market. Roar & Alizadeh (Adland & Alizadeh, 2018), by studying the price differential relationship between Time Charter (TC) freight rates and FFAs contracts, found that TC prices are, on average, higher than FFA contracts due to the requirement of physical delivery in TC contracts, which creates a convenience premium in TC rates. In the case of possible shipping capacity shortages, TC rates tend to rise (Adland & Cullinane, 2005). These studies offer evidence supporting the presence of convenience yield in the shipping market. Therefore, this paper introduces stochastic convenience yield and establishes a dual-factor stochastic model for spot freight rates in the shipping market.

$$\begin{cases} dS = (\mu - \delta)Sdt + \sigma_s Sdz_1 \\ d\delta = \kappa_\delta(\alpha_\delta - \delta)dt + \sigma_\delta dz_2 \\ dz_1 dz_2 = \rho dt \end{cases} \quad (7)$$

where κ_δ , σ_s , σ_δ , μ is the spot freight price drift rate, σ_s is the volatility of spot freight rates, α_δ is the convenience yield of the long-run mean, κ_δ is the rate of adjustment of the mean reversion of the convenience yield, and σ_δ is its volatility. z_1 and z_2 are the corresponding Brownian motion, ρ is the correlation coefficient between spot freight rate and convenience yield.

Given the definition $X = \ln S$, applying Ito's Lemma, the diffusion process of spot freight rate in formula (7) can be expressed as:

$$dX = \left(\mu - \delta - \frac{1}{2}\sigma_s^2\right)dt + \sigma_s dz_1 \quad (8)$$

In addition, let λ_δ denote the market price of convenience return risk, formula (7) can be rewritten under the equivalence harness measure as:

$$\begin{cases} dS = (r - \delta)Sdt + \sigma_s Sdz_1^* \\ d\delta = [\kappa_\delta(\alpha_\delta - \delta) - \lambda_\delta]dt + \sigma_\delta dz_2^* = \kappa_\delta(\alpha_\delta^* - \delta)dt + \sigma_\delta dz_2^* \\ dz_1^* dz_2^* = \rho dt \end{cases} \quad (9)$$

where

$$\alpha_\delta^* = \alpha_\delta - \frac{\lambda_\delta}{\kappa_\delta}$$

Assuming that X , δ obey the joint normal distribution, according to formula (9) (Erb, Lüthi and Otziger, 2014), the state variables X and δ are expressed as

$$\begin{pmatrix} X \\ \delta \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \tilde{\mu}_X(t) \\ \tilde{\mu}_\delta(t) \end{pmatrix}, \begin{pmatrix} \sigma_X^2(t) & \sigma_{X\delta}(t) \\ \sigma_{X\delta}(t) & \sigma_\delta^2(t) \end{pmatrix} \right)$$

where

$$\left\{ \begin{aligned} \tilde{\mu}_X(t) &= X_0 + \left(r - \frac{1}{2}\sigma_s^2 - \alpha_\delta^* \right) t + (\alpha_\delta^* - \delta_0) \frac{1 - \exp(-\kappa_\delta T)}{\kappa_\delta} \\ \tilde{\mu}_\delta(t) &= \exp(-\kappa_\delta T) \delta_0 + \alpha_\delta^* (1 - \exp(-\kappa_\delta T)) \\ \sigma_X^2(t) &= \frac{\sigma_\delta^2}{\kappa_\delta^2} \left(\frac{1 - \exp(-2\kappa_\delta T)}{2\kappa_\delta} - \frac{2(1 - \exp(-\kappa_\delta T))}{\kappa_\delta} + t \right) + 2 \frac{\sigma_s \sigma_\delta \rho}{\kappa_\delta} \left(\frac{1 - \exp(-\kappa_\delta T)}{\kappa_\delta} - t \right) + \sigma_s^2 t \\ \sigma_\delta^2(t) &= \frac{\sigma_\delta^2}{2\kappa_\delta} (1 - \exp(-2\kappa_\delta T)) \\ \sigma_{X\delta}(t) &= \frac{1}{\kappa_\delta} \left\{ \left(\sigma_s \sigma_\delta \rho - \frac{\sigma_\delta^2}{\kappa_\delta} \right) (1 - \exp(-\kappa_\delta T)) + \frac{\sigma_\delta^2}{2\kappa_\delta} (1 - \exp(-2\kappa_\delta T)) \right\} \end{aligned} \right. \quad (10)$$

Bringing formula (10) into formula (4) and taking logarithms on both sides of the equation yields the log forward price closure solution:

$$\ln F(S, \delta, T) = \ln S - \delta \frac{1 - \exp(-\kappa_\delta T)}{\kappa_\delta} + A(T) \quad (11)$$

where

$$A(T) = \left(r - \alpha_\delta^* + \frac{\sigma_\delta^2}{2\kappa_\delta^2} - \frac{\sigma_s \sigma_\delta \rho}{\kappa_\delta} \right) T + \frac{1}{4} \sigma_\delta^2 \frac{1 - \exp(-2\kappa_\delta T)}{\kappa_\delta^3} + \left(\alpha_\delta^* \kappa_\delta + \sigma_s \sigma_\delta \rho - \frac{\sigma_\delta^2}{\kappa_\delta} \right) \frac{1 - \exp(-\kappa_\delta T)}{\kappa_\delta^2} \quad (12)$$

2.3. State-space model

Both single-factor and dual-factor models in the state space framework can estimate the parameters and unobservable state variables through Kalman filtering (Schwartz, 1997). Generally, spot freight and convenience yield are considered unobserved variables, while the FFAs traded in shipping derivatives are observable. Therefore, in empirical research, we use the prices of FFAs with varying maturity dates as observable variables and treat spot freight and convenience yield as state variables. This paper develops a single-factor state space model using formulas (2) and (6), as well as a dual-factor state space model incorporating formulas (7), (8), and (11).

2.3.1. Single-factor state space model

Let $y_t = \ln F_t$, $F_t = [F_t^1, F_t^2, \dots, F_t^5]^T$ is a 5×1 dimensional vectors, and F_t^i denote the i -th FFAs observation at time of t . From formula (6), the observation equation in the state space can be expressed as

$$y_t = d_t + Z_t X_t + \varepsilon_t \quad (13)$$

or expressed as:

$$y_t^i = d_t^i + Z_t^i X_t + \varepsilon_t^i \quad (14)$$

where $y_t^i = \ln F_t^i$, $Z_t^i = e^{-\kappa_\delta T_i}$, $d_t^i = (1 - e^{-\kappa_\delta T_i}) \alpha_\delta^* + \sigma_s^2 (1 - e^{-2\kappa_\delta T_i}) / 4\kappa_\delta$, $i = 1, 2, \dots, 5$, and T_i denotes the maturity date corresponding to the i -th FFAs. The other symbols are as described previously. ε_t is the uncorrelated perturbation term, and $E(\varepsilon_t) = 0$, $\text{Var}(\varepsilon_t) = H$.

By expressing formula (2) in discrete form, the state equation in the

state space can be represented as:

$$X_t = c_t + Q_t X_{t-1} + \eta_t \quad (15)$$

where $c_t = \kappa_\delta \alpha_\delta \Delta t$, $Q_t = 1 - \kappa_\delta \Delta t$, η_t is the uncorrelated perturbation term, $E(\eta_t) = 0$, $\text{Var}(\eta_t) = \sigma_s^2 \Delta t$. The observation formula (13) and the state formula (15) constitute a one-factor state space model.

2.3.2. Two-factor state space model

Similarly, let $y_t = \ln F_t$ and $F_t = [F_t^1, F_t^2, \dots, F_t^5]^T$ be a 5×1 dimensional vector, the observation equation in the state space can be expressed as

$$y_t = d_t + Z_t [X_t \delta_t]^T + \varepsilon_t \quad (16)$$

Or express it as

$$y_t^i = d_t^i + Z_t^i [X_t \delta_t]^T + \varepsilon_t^i \quad (17)$$

where $y_t^i = \ln(F_t^i)$, $Z_t^i = \left[1, -\frac{1 - \exp(-\kappa_\delta T_i)}{\kappa_\delta} \right]$, $d_t = [A(T_i)]$, and $i = 1, 2, \dots, 5$.

T_i denotes the maturity date corresponding to the i -th FFAs. ε_t is the uncorrelated perturbation term, $E(\varepsilon_t) = 0$, $\text{Var}(\varepsilon_t) = H$. Writing the convenience gain diffusion process in formulas (7) and (8) in discrete form, the equation of state in the state space can be expressed as

$$[X_t, \delta_t]^T = c_t + Q_t [X_{t-1}, \delta_{t-1}]^T + \eta_t \quad (18)$$

where $c_t = \left[\left(\mu - \frac{1}{2}\sigma_s^2 \right) \Delta t, \kappa_\delta \alpha_\delta \Delta t \right]^T$, $Q_t = \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 - \kappa_\delta \Delta t \end{bmatrix}$, η_t is the uncorrelated perturbation term, $E(\eta_t) = 0$, $\text{Var}(\eta_t) = \begin{bmatrix} \sigma_s^2 \Delta t & \rho \sigma_s \sigma_\delta \Delta t \\ \rho \sigma_s \sigma_\delta \Delta t & \sigma_\delta^2 \Delta t \end{bmatrix}$. Formula (16) for the observation equation and (18) for the state equation constitute the state space model for a dual factor. This study employs the Kalman filtering method to estimate the state

Table 2
Descriptive Statistics of C5 FFAs.

Contracts	# of Obs.	Min value	Max value	Average value	Standard deviation	Kurtosis	Skewness
M0	1908	2.88	20.50	7.88	2.88	4.63	0.97
M1	1908	2.89	18.88	7.99	2.82	3.60	0.77
M2	1908	2.97	17.00	8.00	2.76	3.01	0.65
Q1	1908	3.25	15.25	7.93	2.63	2.93	0.59
Y1	1908	3.85	11.50	7.55	1.77	2.05	−0.18

Table 3
Parameter Estimation Results of C3 FFA.

Parameters	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	Full sample
Panel A One-factor model												
N	155	202	190	217	224	137	94	188	227	194	83	1908
σ_s	1.95	3.22	0.98	2.44	4.80	0.77	0.85	2.96	2.29	0.90	2.27	0.77
κ_s	3.40	5.76	0.93	3.61	8.34	0.00	0.78	4.02	0.04	0.04	0.73	0.00
α_s^*	1.52	0.01	−0.25	−0.20	0.10	1.90	−0.31	−0.11	1.87	17.84	2.70	2.40
Panel B Two-factor model												
N	155	202	190	217	224	137	94	188	227	194	83	1908
σ_s	0.35	1.04	0.46	0.81	0.30	0.48	0.49	0.61	0.61	0.48	0.74	0.57
σ_δ	0.52	1.37	0.36	2.26	5.54	4.73	0.64	1.73	0.58	0.45	3.28	0.49
κ_δ	0.99	2.40	0.10	2.44	13.73	9.19	0.94	2.63	0.03	0.00	2.45	0.07
ρ	0.78	0.71	0.76	0.89	0.66	0.79	0.91	0.84	0.95	0.95	0.83	0.84
λ_δ	2.93	0.80	0.69	0.63	0.11	1.30	2.65	1.97	0.09	−0.10	0.31	0.28
α_δ^*	−0.47	−0.23	−3.77	−0.51	0.04	0.07	−0.31	−0.12	0.94	736.57	0.43	−1.64

space model.

3. Data

The subject of this paper is the Cape-type FFAs. The sample period covers 1908 trading days, extending eleven consecutive years from January 7th, 2013, to July 5th, 2023.

FFAs are a shipping financial derivative product in which 2 parties, through a shipping brokerage firm, agree to exchange a certain amount of shipping services or shipping costs at a pre-determined price at a future date. Because the shipping market is for transportation services, unlike other common derivatives, FFA contracts can be traded simultaneously in three different terms: monthly, quarterly, and yearly. Among them, monthly contracts include the nearest 5 expiry months starting from the current month, quarterly contracts include the nearest four expiry quarters starting from the next quarter, and yearly contracts include the nearest two expiry years starting from the following year. Therefore, 11 different contracts of the same underlying are traded in the market daily.

Tables 1 and 2 show that contracts with more extended expiration periods have higher minimum and lower maximum values, and vice versa. The standard deviation shows that short-term contracts

demonstrate more splendid volatility, whereas long-term contracts exhibit inferior volatility. These characteristics reflect the mean-reverting nature of FFAs: short-term contracts tend to fluctuate around long-term contracts.

Regarding the mean values, there is an initial increase followed by a subsequent decrease as the expiration period extends, and the Y1 contract demonstrates the lowest mean value. This observation implies the potential presence of non-monotonic characteristics within the term structure of FFAs. Suppose the two-factor model can more satisfactorily fit the non-monotonic features of the FFA term structure. In that case, it is crucial evidence of the effect of convenience yield on the shipping market.

4. Empirical analysis

4.1. Parameter estimation

We employ single-factor and two-factor models to estimate the daily data of FFAs. Schwartz (Schwartz, 1997) pointed out that the parameter estimation results vary with the research period when proposing this type of term structure model. Therefore, it is customary in similar studies to partition the data into distinct sample intervals for estimation

Table 4
Parameter Estimation Results of C5 FFA.

Parameters	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	Full sample
Panel A One-factor model												
N	155	202	190	217	224	137	94	188	227	194	83	1908
σ_s	15.48	4.44	7.61	2.48	3.08	1.62	0.75	2.77	1.41	1.00	3.08	2.38
κ_s	10.83	7.18	5.77	5.09	7.86	2.72	1.28	4.36	1.16	0.11	4.13	2.53
α_s^*	−1.86	0.38	−0.83	−0.12	0.01	0.16	−0.25	−0.06	0.18	6.33	0.31	−0.17
Panel B Two-factor model												
N	155	202	190	217	224	137	94	188	227	194	83	1908
σ_s	2.12	1.71	1.86	0.76	0.43	0.53	0.40	0.69	0.61	0.54	0.63	1.03
σ_δ	2.66	2.35	5.18	2.36	1.91	1.70	0.40	1.93	0.59	0.46	7.52	0.96
κ_δ	1.61	3.74	3.15	5.56	4.71	2.93	0.72	2.83	0.34	0.00	7.14	0.84
ρ	0.70	0.62	0.74	0.64	0.86	0.89	0.81	0.92	0.89	0.93	0.74	0.80
λ_δ	0.70	0.61	−0.21	3.71	0.72	1.27	3.31	1.11	0.36	0.23	0.37	0.38
α_δ^*	−2.38	−0.41	−1.11	−0.14	−0.05	−0.04	−0.55	−0.25	0.55	−547.88	0.18	−0.38

purposes. In the shipping market, a natural year typically constitutes a complete operating cycle. In addition to estimating the entire sample, this paper divides the data into annual intervals for yearly estimation.

Table 3 presents the parameter estimation results for each year of the C3 contract from 2013 to 2023 and the results for the entire sample. N denotes the respective sample size. Panel A presents the parameter estimation results for the single-factor model. The results demonstrate variations in the estimated parameters for spot freight volatility σ_s and mean-reverting speed κ_s among different years. Specifically, the spot freight volatility is relatively high in 6 years, 2014, 2016, 2017, 2020, 2021, and 2023, with 2017 exhibiting the highest σ_s and κ_s . A tremendous value of α_s^* in 2022 reveals a significant increase in freight rates during COVID-19.

Panel B is the parameter estimation result of the two-factor model. The empirical value of 0.03 is assigned to the market interest rate during the estimation process. The results indicate that 2013, 2014, 2016, 2017, 2018, 2019, 2020, and 2023 exhibit high levels of convenience yield volatility. From 2013 to 2023, the spot freight volatility decreased compared to the single-factor model in Panel A, indicating that the convenience yield volatility explains part of the spot freight volatility in the two-factor model. The convenience yield exhibits a relatively high mean-reverting speed, demonstrating the significant presence of mean-reversion in the market. This phenomenon is particularly evident in years with high convenience yield volatility (such as 2017, 2018, and 2023). The estimated correlation coefficients ρ are all above 0.66, delivering evidence of a robust positive correlation between convenience yield and spot freight, which aligns with observed trends in

commodity markets.

Table 4 presents the parameter estimation results for each year of the C5 contract from 2013 to 2023 and the results for the entire sample. The results demonstrate variations in the estimated parameters for spot freight volatility σ_s and mean-reverting speed κ_s among different years. Specifically, the spot freight volatility is relatively high in five years, 2013, 2014, 2015, 2017, and 2023, with 2013 exhibiting the highest mean-reverting speed κ_s .

Table 4 shows the parameter estimation results for each year of the C5 contract from 2013 to 2023 and the results for the entire sample. The results demonstrate variations in the estimated parameters for spot freight volatility σ_s and mean-reverting speed κ_s among different years. Notably, the spot freight volatility is relatively high in five years, 2013, 2014, 2015, 2017, and 2023, with 2013 displaying the most elevated σ_s and κ_s .

Panel B is the parameter estimation result of the two-factor model. The empirical value of 0.03 is also assigned to the market interest rate. The results indicate that 2013, 2014, 2015, 2016, 2017, 2018, 2020, and 2023 exhibit elevated levels of convenience yield volatility. σ_δ shows an insignificant impact in 2019, 2021, and 2022, but the values are close to the values of σ_s . Table 3 of the C3 shows the same results in 2019, 2020, and 2022. The considerable increase in freight rate during COVID-19 dilutes the impact of convenience yield. From 2013 to 2023, the spot freight volatility significantly decreased compared to the single-factor model, indicating that in the two-factor model, part of the spot freight volatility is explained by the convenience yield volatility. The convenience yield exhibits a relatively high mean-reverting speed,

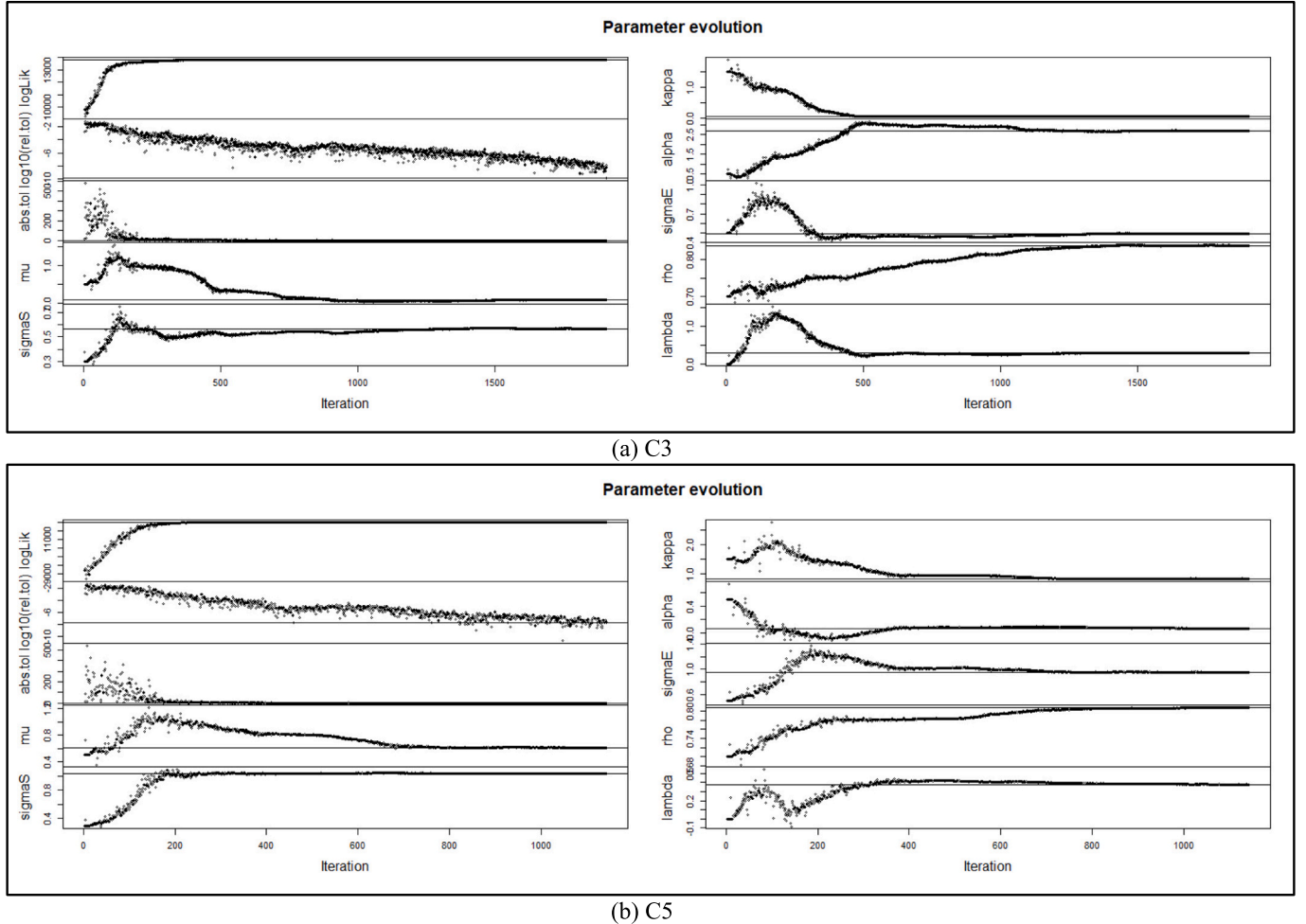


Fig. 1. Parameter Estimation Iteration Process.

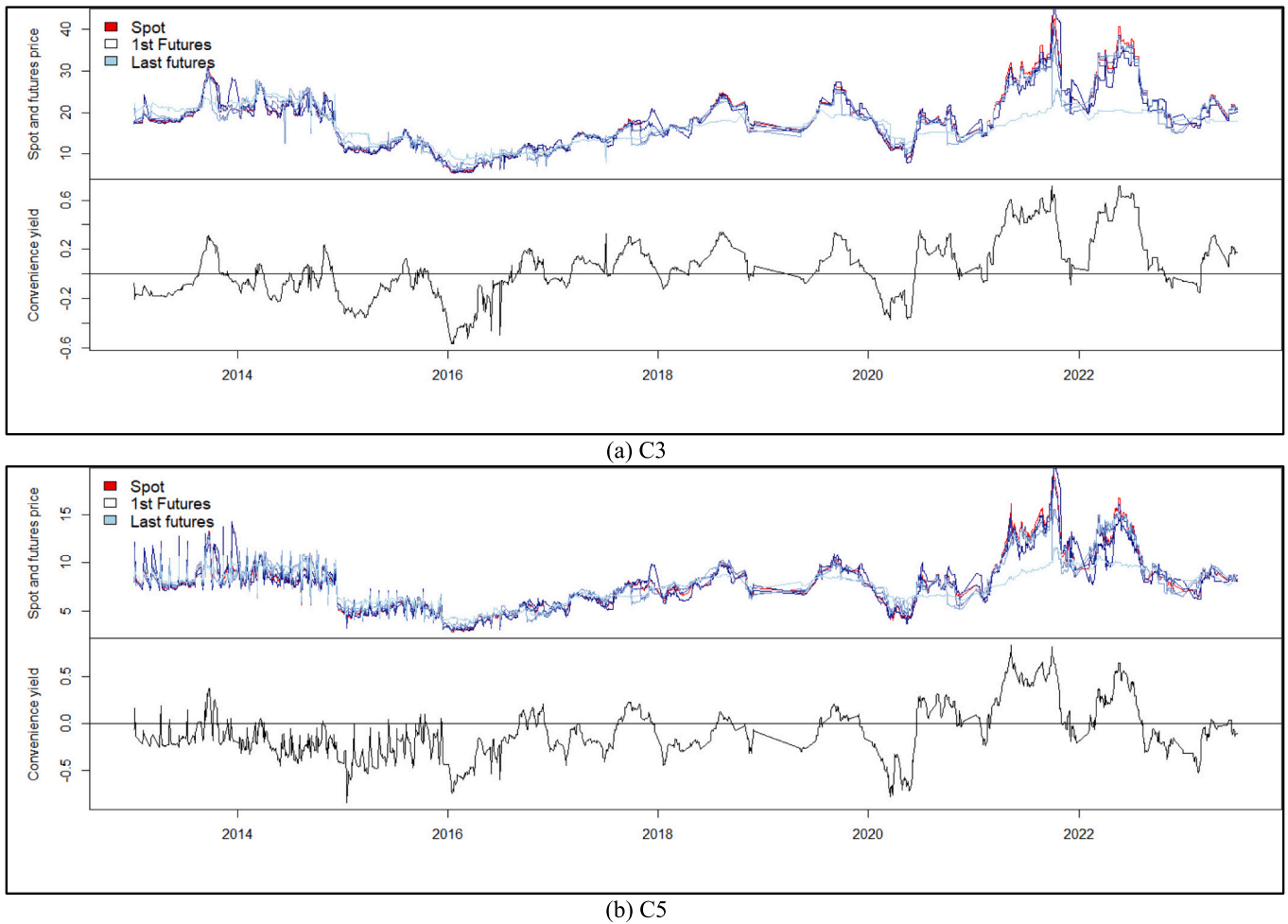


Fig. 2. Capesize Vessel State Variable Fitting.

demonstrating the significant presence of mean-reversion in the market. This phenomenon is particularly evident in years with high convenience yield volatility. The estimated correlation coefficients are all above 0.62, providing evidence of a robust positive correlation between convenience yield and spot freight, which aligns with observed trends in commodity markets.

Fig. 1 illustrates the parameter estimation iterations of C3 and C5, which show all parameters obtain convergence and the estimated values are convincing.

4.2. Term structure

Under the dual-factor model, convenience yield can be estimated as a state variable. Fig. 2 shows the estimated convenience yield and spot freight and FFAs price series for each C3 route year at the top and the corresponding data results for the C5 route at the bottom.

Fig. 2 illustrates a significant correlation between the trends of convenience yield and spot freight rates. This finding is consistent with the estimated correlation coefficient results in Tables 3 and 4.

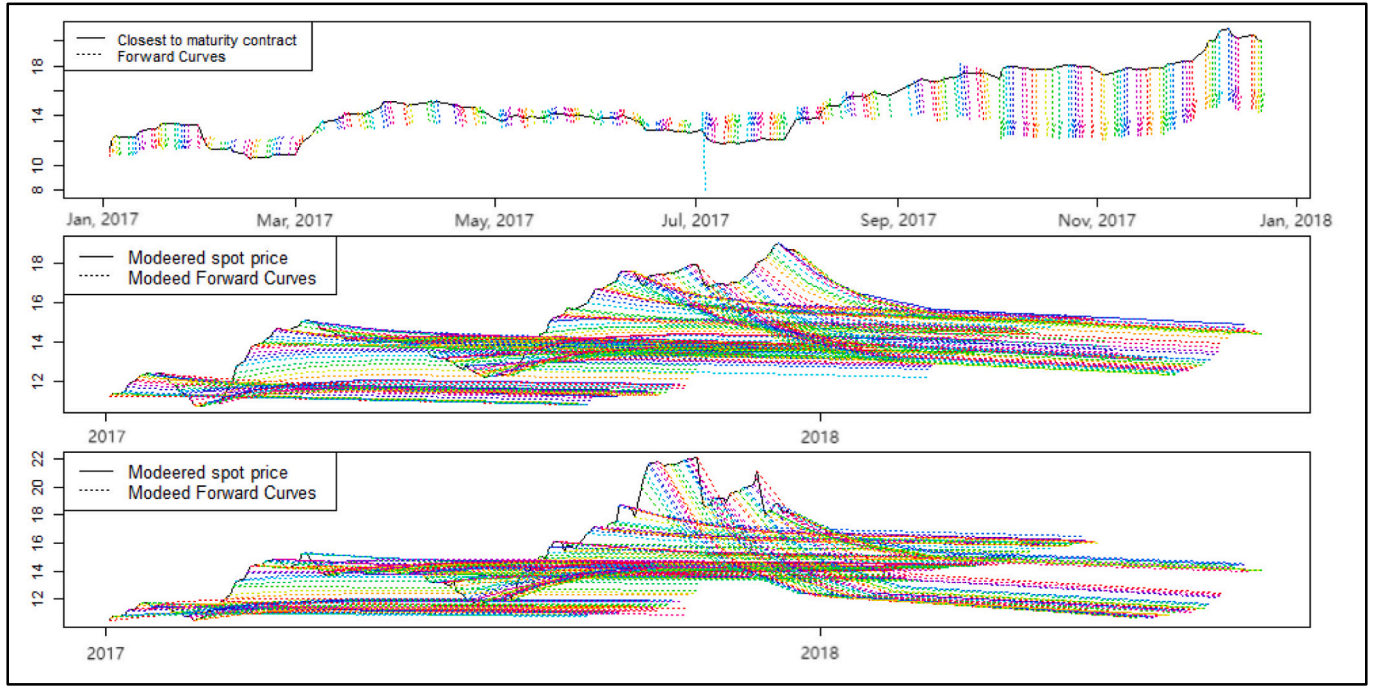
Additionally, the shipping market exhibits evident seasonality and cyclicity. Seasonal fluctuations in dry bulk shipping demand arise from various factors, including increased coal demand during the winter heating season, cyclicity in agricultural production, and the traditional holidays. Furthermore, the prolonged off-season period in the transportation market is caused by year-end shutdowns and maintenance work, leading to reduced demand for bulk commodities. Consequently, the international shipping market designates the first and second quarters as low seasons. However, the third and fourth quarters are

recognized as high seasons due to these factors. This seasonality is reflected in the convenience yield. The convenience yield is generally low or negative during the first half of most years (Kavussanos & Visvikis, 2006), while the highest convenience yield is observed from July to October. This is because the transportation of bulk commodities primarily dictates the convenience yield in the shipping market. Fluctuations in the convenience yield of bulk commodities are caused by factors such as demand and supply, which in turn influence the international maritime market and lead to changes in the convenience yield in the shipping market.

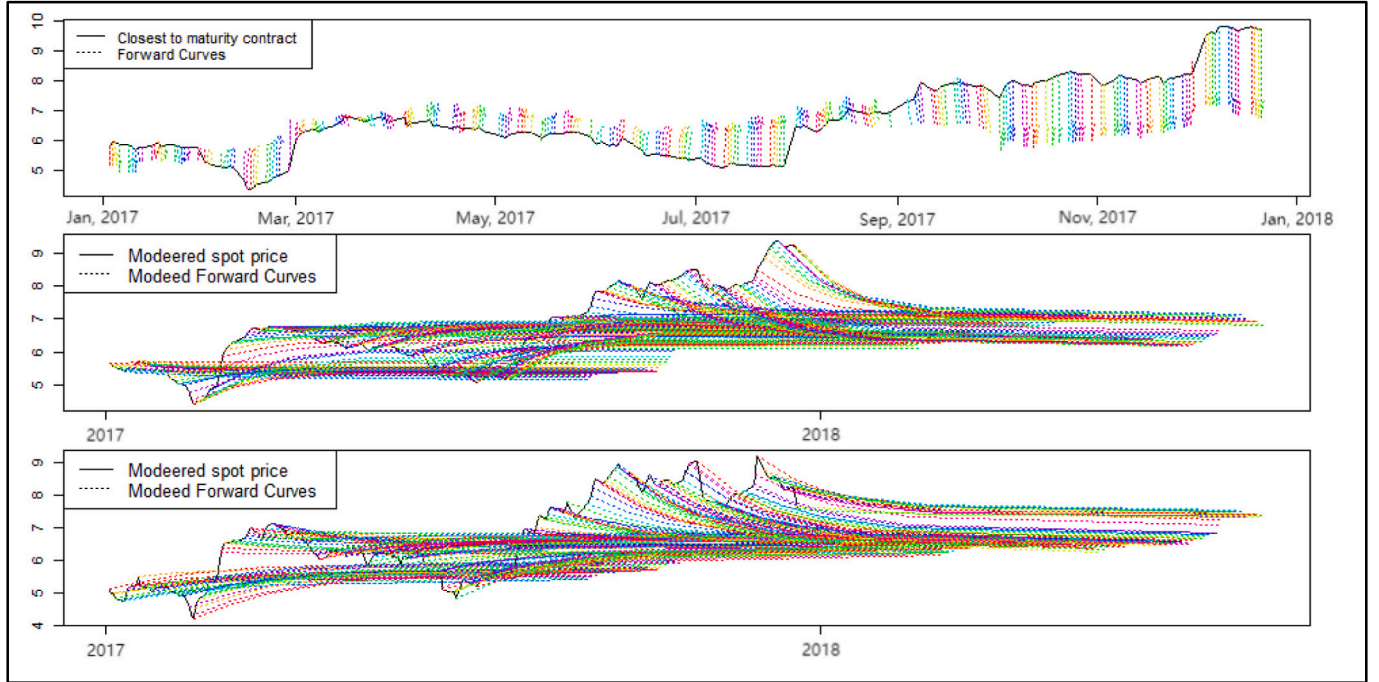
Lastly, the convenience yield series is associated with variations in the term structure of FFAs. By comparing the convenience yield at the bottom and the spot freight rates & FFA price sequences at the top of each subfigure in Fig. 2, it can be observed that when the convenience yield is high, the spot freight rates are generally higher than the FFA prices. Conversely, during periods of low convenience yield, the spot freight rates mainly fall below the FFA curve.

Fig. 3 illustrates the results of fitting the 2017 FFA term structure for routes C3 and C5 to compare the two models further. Fig. 3 displays, from top to bottom, the real term structure of FFA and the fitted results obtained from the single and dual-factor models. The comparison in Fig. 3 reveals that the dual-factor model performs more reasonably than the single-factor model in fitting the term structure. The results are even more pronounced for route C5, where the one-factor model performance could be more satisfactory when fitting the non-monotonic variation characteristic of the FFA term structure.

Based on formula (6), the FFAs price converges to a value as the contract expiration approaches infinity:



(a) C3



(b) C5

Fig. 3. FFAs Term Structure.

$$F(S, \infty) = \exp \left[\alpha_s^* + \frac{\sigma_s^2}{4\kappa_s} \right] \quad (19)$$

In the single-factor model, it is undeniable that whether FFAs are in contango or backward depends on the relationship between spot freight rates and the convergence price. If spot freight rates exceed the convergence price, forward prices are in backwardation; conversely, if spot freight rates are below the convergence price, forward prices are in contango. The strict assumptions of the single-factor model limit its ability to fit the term structure effectively. This is due to the single-factor model's sole consideration of spot freight rates as a state variable,

resulting in the omission of significant market information.

In contrast, the dual-factor model assumes that the convenience yield follows a mean-reverting process. Lüthi et al. (Lüthi, Erb, & Otziger, 2014) pointed out that the relationship among risk-free interest rates r , convenience yield δ_t , and the long-term average of convenience yield α_δ^* determines the shape characteristics of the term structure. According to formula (9), when $r > \delta_t$, the drift rate of spot freight rates was positive, meaning that freight prices would drift upward in the short term. At this time, the term structure will exhibit a near-end upward slope. When $r < \delta_t$, the drift rate of spot freight rates is negative, leading to a downward drift in freight prices in the short term. The term structure

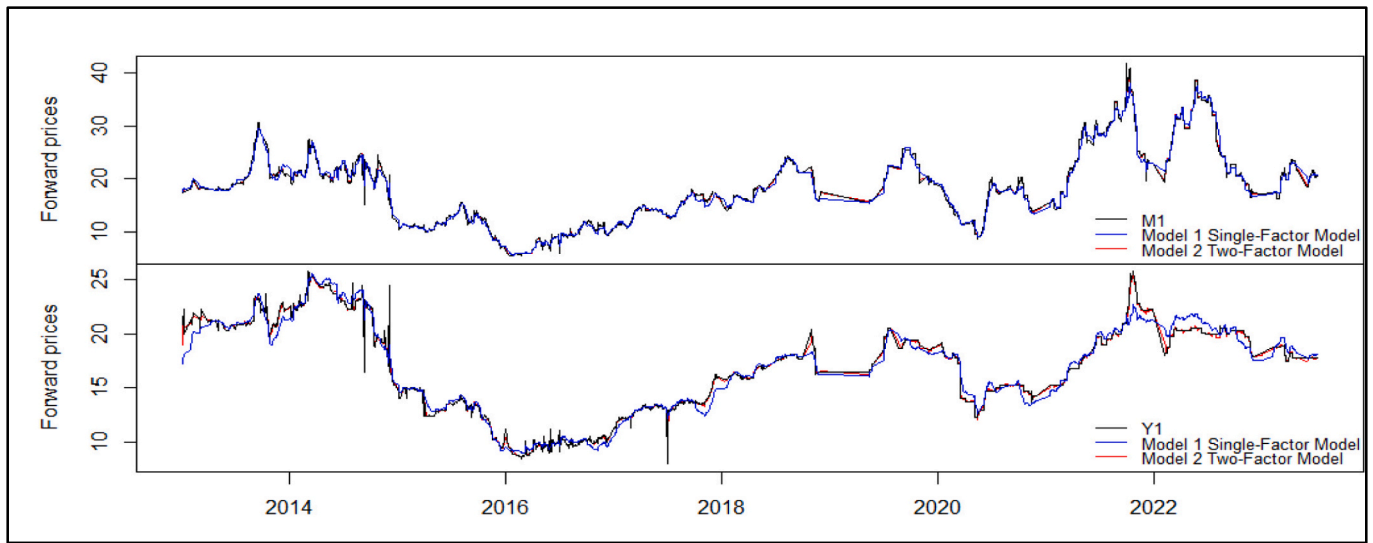
Table 5
Term Structure Morphology.

Serial number	Relationships	Term structure pattern	Forward FFAs contango and backwardation
1	$r > \delta_t$ and $r > \alpha_\delta^*$	Monotonically increasing	Contango increases
2	$r > \delta_t$ and $r < \alpha_\delta^*$	Increasing then decreasing	Contango initially increases and then decreases
3	$r < \delta_t$ and $r > \alpha_\delta^*$	Decreasing then increasing	Backwardation initially increases and then decreases
4	$r < \delta_t$ and $r < \alpha_\delta^*$	Monotonically decreasing	Backwardation increases

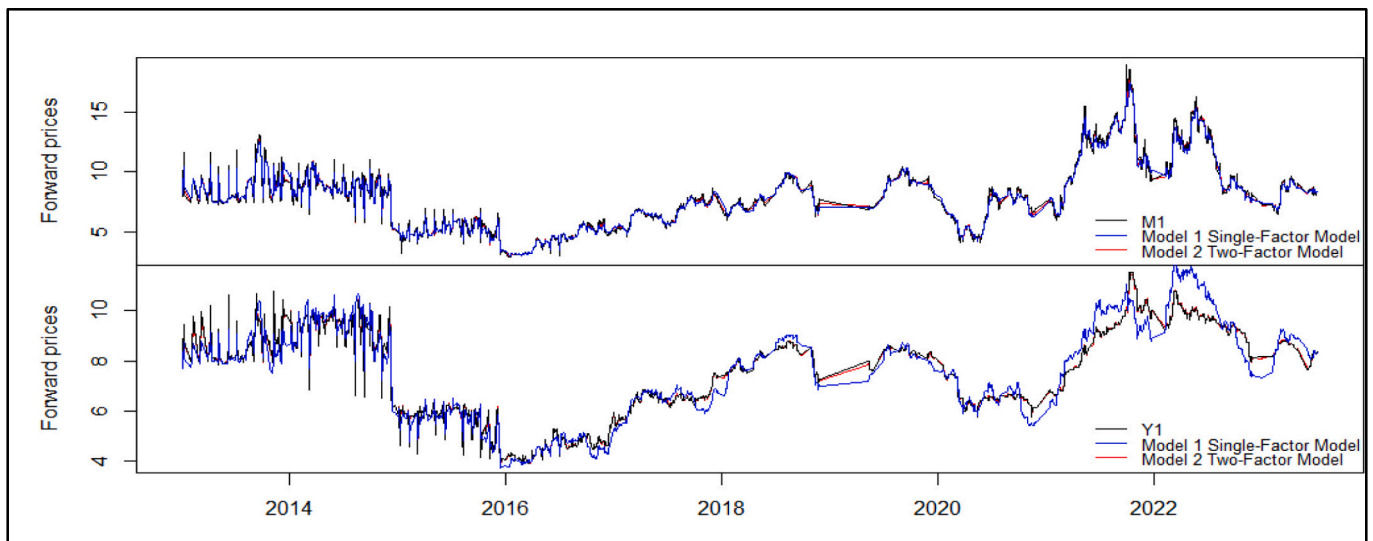
will exhibit a downward slope in the near end. Therefore, if $r > \alpha_\delta^*$, then it implies that the future convenience yield δ_t remains less than r , in the long term, freight prices will drift upwards. At this time, the term structure will exhibit an upward slope in the far end. On the contrary, $r < \alpha_\delta^*$ implies that the future convenience yield δ_t is more significant

than r , and the term structure will exhibit a downward slope in the far end. Table 5 presents the four types of term structure patterns for FFAs in the bivariate model.

The route C5 convenience yields remained relatively low from January to July 2017 (see Fig. 3), when $r > \delta_t$, where r is set as 0.03, the market is at a low level, and investors expect freight prices to rise in the short term. Therefore, the near-end term structure exhibits significant FFA premiums. Since the long-term average α_δ^* of δ_t is -0.05 (see Table 4 Panel B), at this time $r > \alpha_\delta^*$, investors anticipate a positive long-term market. Therefore, the term structure exhibits the first pattern, with increasing long-term FFA premiums and an upward slope in the far-end term structure (see Fig. 3). In July 2017, the shipping market entered the peak season, with the market continuing to rise and remaining high from July to November. At this time $\delta_t > r$, investors expected that freight prices might decrease in the short term, resulting in a short-term FFAs discount and a downward sloping near-term term structure. Because of $r > \alpha_\delta^*$, the term structure in the distant end exhibits an upward sloping curve. Overall, during this period, the term structure exhibited a third



(a) C3



(b) C5

Fig. 4. FFAs Curve Fitting.

Table 6
C3 Fitting Results.

Expiration date	MAE (of Log Prices)		MSE (of Log Prices)	
	One-factor model	Two-factor model	One-factor model	Two-factor model
M0	0.0780	0.0606	0.0100	0.0067
M1	0.0587	0.0033	0.0053	0.0010
M2	0.0544	0.0400	0.0048	0.0033
Q1	0.0700	0.0787	0.0087	0.0116
Y1	0.1100	0.0048	0.0214	0.0001
Average value	0.0742	0.0375	0.0101	0.0043

Table 7
C5 Fitting Results.

Expiration date	MAE (of Log Prices)		MSE (of Log Prices)	
	One-factor model	Two-factor model	One-factor model	Two-factor model
M0	0.0865	0.0647	0.0113	0.0075
M1	0.0514	0.0143	0.0041	0.0004
M2	0.0464	0.0428	0.0034	0.0032
Q1	0.0673	0.0724	0.0075	0.0010
Y1	0.0881	0.0033	0.0137	0.0010
Average value	0.0679	0.0395	0.0080	0.0042

form, namely a decline followed by an increase (see Fig. 3).

According to Table 4 Panel B, the spot freight of C5 volatility σ_s in 2021 is 0.61, and the convenience yield σ_δ volatility is 0.59. This indicates a lower overall market volatility compared to 2017. However, the long-term average of the convenience yield α_δ^* in 2021 is 0.55, which means $r < \alpha_\delta^*$. Therefore, from January to July 2021, when $r > \delta_t$, the market is at a low level, the term structure exhibits the second form, characterized by an initial increase followed by a decrease. From July to November 2021, when $\delta_t > r$, the market is at a high level and investors anticipate a decrease in short-term freight rates, the term structure assumes the fourth form, a monotonic downward slope.

Based on the above analysis, it is evident that the two-factor model, which introduces random convenience yield, outperforms the single-factor model in fitting the FFA term structure. This finding supports the existence of a convenience yield in the shipping market and its ability to provide valuable information. Additionally, the stochastic variation of the convenience yield plays a significant role in shaping various FFA term structure forms.

4.3. Fitting effectiveness

Fig. 4 compares routes C3 and C5 fitting by selecting the FFA M1 for shorter terms and Y1 for longer terms. The curves fitted by the two models and the actual FFA trend are close by comparing the fitting of the single and two-factor models. However, the curves under the two-factor model are more comparative to the actual FFA prices. The curve produced by the two-factor model exhibits even closer alignment with the actual FFAs prices, primarily notable for the one-year FFAs, where the curve generated by the two-factor model almost perfectly overlaps with the actual FFAs curve.

Table 8
Descriptive Statistics of r_c , δ_t , and $\Delta\delta_t$ of C3 Freight Index.

Variables	Quantity	Minimum	Maximum	Average	Range	Standard deviation	Skewness	Kurtosis
r_c	1903	-4.5326	1.2852	-0.0020	5.8178	0.1501	-14.6201	454.7188
δ_t	1903	-0.5657	0.7147	0.0698	1.2804	0.2393	0.3406	0.0862
$\Delta\delta_t$	1903	-0.4359	0.3165	0.0001	0.7525	0.0325	-0.5678	33.4315

Tables 6 and 7 present the Mean Absolute Error (MAE) and Mean Squared Error (MSE) of the single-factor and two-factor models for routes C3 and C4, respectively. The bold indicates the best results. These metrics evaluate the effectiveness of the models in fitting FFA prices across various expiration periods. As seen from Tables 6 and 7, there are slight differences in the fitting performance for FFA contracts with different terms. Overall, the two-factor model yields lower average MAE and MSE than the single-factor model. Notably, the two-factor model exhibits significantly superior fitting performance for shorter-term contracts, such as M0 and M1, and for longer-term contracts, Y1. However, for M2 and Q1, the two-factor models perform similarly to the one-factor one, even worse, indicating that the convenience yields are more substantial in short and long-term FFA, not in medium ones.

In summary, the two-factor model demonstrates the capacity to capture the various trends within the FFA term structure. It effectively showcases the non-linear attributes of the term structure and can accurately fit diverse market conditions. Compared to the single-factor model, the two-factor one displays the most excellent fitting capabilities for the FFA term structure. This further corroborates the presence of convenience yield in the shipping market, which encapsulates substantial information about the industry.

5. Convenience yield for market forecasting

The term structure integrates available public information and participants' future expectations, offering valuable insights for hedging, investment decisions, and risk management in the spot market (Lautier, 2005). As a critical factor affecting the term structure, convenience yield plays a significant role. This section further explores the predictive role of convenience yield (and the market information it encompasses) in the freight market.

5.1. Shipping market

The FFAs for Capesize vessels are linked to the BCI, which reflects the spot freight market trends specific to Capesize vessels in this study. The BCI data spans six years, from January 7th, 2013, to July 5th, 2023, obtained from the Baltic Exchange, a prominent shipping exchange. Additionally, the two-factor model estimates the convenience yield in the shipping market. To ensure accurate predictions, a rolling window estimation technique is employed for the convenience yield δ_t , eliminating the use of future data. The window length is set at one year, and all indicators are derived from daily data. r_c represents the first-order logarithmic difference of the BCI, while the variable $\Delta\delta_t$ denotes the first-order difference of the convenience yield, which reflects changes caused by the most recent market information. Tables 8 and 9 provide the descriptive analysis of the logarithmic returns of the freight index, convenience yield, and their differences. Tables 10 and 11 present the corresponding correlation results.

The ADF test values of r_c , δ_t , and $\Delta\delta_t$ for C3 are 0.000, 0.0364, and 0.000, respectively; the values for C5 are 0.000, 0.0013, and 0.000, respectively. The results show that all series are stationary. The standard deviations of r_c , δ_t , and $\Delta\delta_t$ are significantly greater than their means, indicating substantial fluctuations in freight rates for the three types of ships.

Table 9Descriptive Statistics of r_c , δ_t , and $\Delta\delta_t$ of C5 Freight Index.

Variables	Quantity	Minimum	Maximum	Average	Range	Standard deviation	Skewness	Kurtosis
r_c	1903	-4.5326	1.2852	-0.0020	5.8178	0.1501	-14.6201	454.7188
δ_t	1903	-0.8428	0.8309	-0.0732	1.6737	0.2823	0.5145	0.2112
$\Delta\delta_t$	1903	-0.4347	0.3735	0.0000	0.8083	0.0606	0.2434	10.2817

Table 10Correlation Analysis of r_c , δ_t , and $\Delta\delta_t$ of C3 Freight Index.

Variables	r_c	δ_t	$\Delta\delta_t$
r_c	1.000		
δ_t	0.024	1.000	
$\Delta\delta_t$	0.185	0.064	1.000

Table 11Correlation Analysis of r_c , δ_t , and $\Delta\delta_t$ of C5 Freight Index.

Variables	r_c	δ_t	$\Delta\delta_t$
r_c	1.000		
δ_t	0.039	1.000	
$\Delta\delta_t$	0.137	0.108	1.000

5.2. Forecasted results

We considered various models, such as the linear model (ARMAX), non-linear model (GARCH-SK), and machine learning model (Convolution Neural Network, CNN), for fitting and forecasting the data. Each model has its characteristics. Compared to other models, the GARCH-SK model extends the traditional GARCH model and is sufficiently suited to capturing the volatility clustering, skewness, and kurtosis in financial time series data. However, it primarily focuses on the time series' volatility patterns and lacks a direct mechanism to include external variables, making it unsuitable for studying convenience yield, an external variable, in predicting the shipping market. CNNs, initially designed for image processing and computer vision tasks, have been increasingly applied in financial data analysis due to their powerful feature extraction capabilities, especially in handling time series data. Although CNNs have unique advantages in fitting and forecasting financial data, their black-box nature, poor interpretability, and substantial data requirements for training make them less adaptable to time series data characteristic of the shipping market. The ARMAX model, a standard method in time series analysis, can consider both endogenous variables within the time series and exogenous variables, enhancing the accuracy and reliability of forecasts. Compared to many black-box machine learning models, the parameters of the ARMAX model have clear economic implications, making it more suitable for our study.

Before constructing the ARMAX model, we verified the stationarity of the data and utilized the Granger causality test to ensure the exogeneity of the added exogenous variables. The results are as follows: The

Table 12

C3 ARMAX Model Results.

	(1)	(2)	(3)
δ_t		-0.005 (0.868)	
$\Delta\delta_t$			0.303*** (0.004)
AR(-1)	0.108 (0.145)	0.108 (0.164)	0.070 (0.382)
MA(-1)	0.078* (0.086)	0.128 (0.102)	0.153* (0.059)
L-L	941	941	945
AIC	-1873	-1871	-1880
BIC	-1852	-1844	-1852

Table 13

C5 ARMAX Model Results.

	(1)	(2)	(3)
δ_t		0.023 (0.315)	
$\Delta\delta_t$			0.299*** (0.00)
AR(-1)	0.115 (0.215)	0.112 (0.237)	0.049 (0.622)
MA(-1)	0.078 (0.412)	0.080 (0.315)	0.136 (0.178)
L-L	924	925	938
AIC	-1839	-1840	-1867
BIC	-1817	-1812	-1839

Note: The values in parentheses '()' indicate the p-values. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

logarithmic returns of the freight index pass the ADF stationarity test (ADF statistic: -4.1458, p -value: 0.0008). For the C3 route, the test result for the convenience yield is $F = 1.4726$, $p = 0.0479$, and for the first difference of the convenience yield, $F = 1.6041$, $p = 0.0205$. Since both p -values are less than 0.05, we can consider them exogenous. Similarly, for the C5 route, the test result for the convenience yield is $F = 2.0111$, $p = 0.0201$, and for its first difference, $F = 1.9853$, $p = 0.0221$. These can also be considered exogenous.

The ARMAX (1,1) model is constructed for the logarithmic returns of the freight index, with convenience yield and its difference sequence taken as exogenous variables. To analyze the predictive role of convenience yield, the exogenous variables incorporate lagged data from one period. The ARMAX (1, 1) model is defined as follows:

$$y_t = \varphi y_{t-1} + \theta \varepsilon_{t-1} + v x_{t-1} + \varepsilon_t \quad (20)$$

where y_t and y_{t-1} represent the logarithmic returns for the t and $t-1$ periods of the BCI, respectively. y_{t-1} denotes the autoregressive (AR) component of the model, ε_{t-1} represents the moving average (MA) component, and x_{t-1} is the lagged exogenous variable component (including convenience yield and its first-order difference). ε_t represents the model's residuals, and φ , θ , v correspond to the regression coefficients.

Tables 12 and 13 display the forecast results of the ARMAX model. Regression results (2) and (3) show the outcomes of the models when convenience yield and its difference sequence are used as individual exogenous variables, respectively. This study examines the significance, direction, and magnitude of the coefficients of the exogenous variables to investigate the explanatory and predictive role of convenience yield in the spot freight market.

The results in Tables 12 and 13 indicate a significant and positive coefficient for the coefficient of convenience yield difference. In the commodity market, higher inventory levels correspond to lower convenience yield and vice versa. However, unlike commodities, the shipping market's convenience yield is determined by the level of idle capacity. A decrease in idle capacity raises the shipping market's convenience yield and vice versa. The growing demand for commodity transportation results in a reduction of idle capacity, thereby driving up the convenience yield. Casassus & Collin-Dufresne (Casassus & Collin-Dufresne, 2005) indicated that convenience yield contributes to the mean-reverting nature of spot prices in the commodity market. Our research expands upon their work and explores its application in the

shipping market.

The difference in convenience yield reflects changes caused by the most recent market information. A rise in convenience yield within the shipping market indicates a decrease in idle capacity, leading to an upsurge in freight prices. Our research establishes the predictive capabilities of convenience yield in the shipping market for spot freight prices. This development builds upon the findings of Gospodinov & Ng (Gospodinov & Ng, 2013), who demonstrated the predictive nature of convenience yield in commodity prices.

6. Conclusions

6.1. Conclusions of study

Risk management is crucial in the shipping market, and FFAs are a significant hedging mechanism for freight risk. This empirical study explores the convenience yield in the shipping market using FFA data from 2013 to 2023. Based on the mean-reverting characteristics of the shipping market, this research applies classic random process models commonly used in commodity markets. A comparative analysis is conducted between the single-factor model, which solely considers the mean reversion of spot freight and the two-factor model. The latter incorporates the mean-reverting process of convenience yield to fit the term structure of FFAs. The findings are as follows:

Firstly, at the data level, the results support the presence of convenience yield in the shipping market as indicated by the two-factor model. Including the stochastic process of convenience yield, the two-factor model outperforms the single-factor one in fitting the term structure by effectively capturing the non-monotonic characteristics observed in the FFAs term structure. The stochastic fluctuation of convenience yield emerges as a crucial factor influencing the various forms of term structures in FFAs.

Secondly, the convenience yield in the shipping market is influenced by the commodities being transported. The literature review indicates that the convenience yield is an endogenous outcome of the interactions among supply, demand, and storage decisions. Our study demonstrates the convenience yield within the shipping market, particularly in transporting bulk commodities. Therefore, for markets where storage is not feasible, variables associated with these markets can facilitate the transfer of convenience yields through production and consumption processes. Integrating the dynamic correlation between shipping and commodity markets, it is discernible that convenience yields can be transferred from commodity markets to shipping markets, thereby extending the research on convenience yields to non-storage shipping sectors. Consequently, the convenience yields in the shipping market stem from transporting bulk commodities. Further, the convenience yield in the shipping market might be linked to idle shipping capacity. An increase in the demand for bulk commodity transportation reduces idle shipping capacity, elevating the convenience yield in the shipping market. The lesser the idle capacity, the higher the convenience yield in the shipping market, and vice versa. Factors such as demand and supply can increase or decrease the convenience yield of bulk commodities, affecting the international shipping market and resulting in changes to the convenience yield in the shipping market. However, at present, we only ascertain that the supply and demand factors of bulk commodities influence the shipping market's convenience yield, but the specific transmission mechanisms still need to be determined. This represents a direction for future research. Additionally, during the COVID-19 period of 2020–2021, the global shipping supply chain experienced significant disruptions, leading to a notable decline in shipping capacity. Based on the deduction above, the increased demand for bulk commodity transportation consequently elevated the convenience yield in the shipping market, pushing freight prices higher. This aligns with the results in Table 3, which documents the considerable increase in freight rates during COVID-19. Also, the convenience yield in the shipping market exhibits clear seasonality and cyclical patterns due to the cyclicity

found in commodity consumption and transportation demands.

Thirdly, the convenience yield and its difference sequence demonstrate significant predictive power for shipping freight rates. The idle capacity in the shipping market determines the convenience yield, enabling it and its difference sequence to explain the price changes of future spot freight rates. These findings contribute to implementing effective risk management strategies in the shipping market.

6.2. Implications for risk Management in the Shipping Market

This study analyzes FFA data for C3 and C5 routes and provides the first empirical evidence of convenience yield's existence and predictive power in the shipping market. These findings have significant implications for risk management in the shipping industry:

6.2.1. Improved forecasting

The predictive power of convenience yield for spot freight rates provides market participants with a powerful tool. Shipping companies, charterers, and investors can develop more accurate predictive models by incorporating convenience yield into forecasting models. This enhanced predictability can directly translate into better-informed decisions on fleet deployment, contract negotiations, and hedging strategies. For instance, when a rise in freight rates is predicted, shipping companies can proactively adjust their pricing strategies or increase capacity.

6.2.2. Early warning system

Fluctuations in convenience yield can serve as early signals of market changes. Our research shows that changes in convenience yield often precede changes in spot freight rates. This characteristic makes convenience yield an effective early warning system. Market participants can closely monitor changes in convenience yield to anticipate potential market turning points. For example, a sudden increase in convenience yield might indicate an imminent shortage in shipping capacity, allowing companies to make strategic adjustments in advance.

6.2.3. Optimal hedging strategies

Understanding the relationship between convenience yield and spot freight rates can help design more effective hedging strategies. Our findings suggest that when convenience yield is high, it may indicate future increases in freight rates. In such situations, shipping companies might lean towards increasing the use of physical assets (ships) rather than relying solely on financial derivatives for hedging. This dynamic hedging strategy based on convenience yield can significantly improve the effectiveness of risk management.

6.2.4. Investment decision support

Convenience yield provides essential reference points for investment decisions in the shipping market. High convenience yield may indicate strong demand for shipping capacity, potentially signaling an excellent time to expand fleet size. Conversely, persistently low convenience yield might suggest market overcapacity, indicating a need to consider fleet reduction or postponement of new ship orders. Investors and shipping companies can better navigate market cycles and optimize long-term investment strategies by focusing on convenience yield.

6.2.5. Risk assessment optimization

Incorporating convenience yield into risk assessment models can provide a more comprehensive view of market risks. Traditional risk assessment methods may focus on price volatility while overlooking the market information in convenience yield. Our research demonstrates that convenience yield captures market supply-demand conditions and future expectations, which is crucial for accurate risk assessment. Integrating convenience yield can achieve more accurate Value-at-risk (VaR) calculations and rational capital allocation decisions.

6.2.6. Improved contract pricing

Understanding convenience yield can significantly improve the pricing of long-term shipping contracts and FFAs. Our findings indicate that convenience yield contains essential information about future market conditions. Incorporating this information into pricing models can achieve more accurate and fair contract pricing, reducing the risk of entering unfavorable contracts. This is significant for shipping companies and charterers, potentially fostering more stable and mutually beneficial long-term partnerships.

This study opens new avenues for research on convenience yield in the shipping market. Further studies could explore the characteristics of convenience yield across different routes and vessel types and the impact of macroeconomic factors on convenience yield. Moreover, developing real-time risk management systems based on convenience yield would be a valuable research direction, providing the shipping industry with more advanced risk management tools. This study confirms the existence of convenience yield in the shipping market and reveals its crucial role in risk management. By profoundly understanding and effectively utilizing the market information in convenience yield, shipping industry participants can significantly enhance their risk management capabilities, strengthening their competitiveness and resilience in this highly volatile market.

In addition to C3 and C5 FFAs, we also conduct studies on C4, C7, and 4TC FFAs. The proposed model performs well in these supplemental routes, and the convenience yields are also significant. Therefore, although the shipping market deals with “non-storable” services, the convenience yield from the commodity market can be transmitted to the shipping market. Convenience yield plays a vital role in determining the characteristics of FFA term structures and in forecasting the shipping market. This study provides a comprehensive understanding of the economic implications and mechanisms of convenience yield in the shipping market, offering new insights for future studies in this field.

CRedit authorship contribution statement

Wenyang Wang: Writing – original draft, Software, Project administration, Methodology, Funding acquisition, Conceptualization, Writing – review & editing. **Zihao Wang:** Writing – original draft, Visualization, Software, Methodology, Data curation, Writing – review & editing. **Fangyi Zhou:** Visualization, Writing – review & editing. **Jinghan Wang:** Software, Visualization, Writing – review & editing. **Jinglin Wang:** Writing – original draft, Software, Data curation, Writing – review & editing. **Cong Sui:** Supervision, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the Humanities and Social Sciences Youth Foundation, Ministry of Education of the People's Republic of China (22YJC910011), the Postdoctoral Research foundation of China (2023M733444), and the National Natural Science Foundation of China (72371045).

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